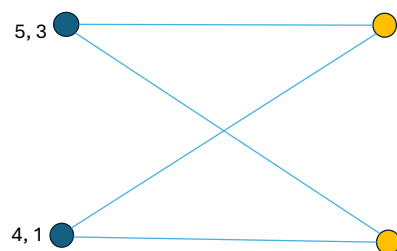
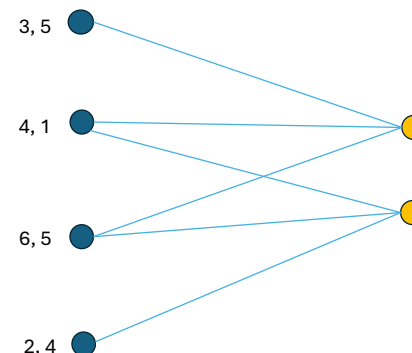




**Example 2:** This is an example with two buyers and two sellers who are fully connected. The  $i$ th number is each buyer's valuation of the  $i$ th good.



**Example 3:** Duopoly sellers with partially-overlapping buyers.



## Model

- **Sellers:**  $S = \{s_1, \dots, s_M\}$ . Each seller sells only one good.
- **Buyers:**  $B = \{b_1, \dots, b_N\}$ . Each buyer needs at most one good.
- **Bipartite Graph:**  $G = \langle S \cup B, E \rangle$  (Definition of bipartite graph: every (undirected) link connects a buyer and a seller.)
- **Value:** For  $(s_i, b_j) \in E$ , buyer  $b_j$ 's value of good  $i$  sold by seller  $s_i$  is  $v_{ij} > 0$ .
- We can interpret no link as there being no valuable economic opportunity. So, we assume that every  $s_i \in S$  and  $b_j \in B$  is connected by assigning  $v_{ij} = 0$  for any  $s_i, b_j$  such that  $(s_i, b_j) \notin E$ .
- We assume  $M = N$  without loss of generality (if  $M \neq N$ , then we add dummy sellers or dummy buyers with 0 values).

## Matching

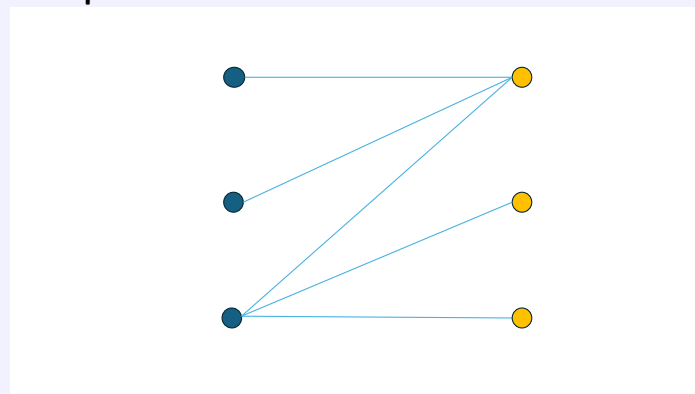
- **Perfect matching:** Each buyer is matched with exactly one seller and each seller is matched with exactly one buyer.
  - ▶ Perfect matching corresponds to the bipartite graph where the degree of every buyer/seller is 1.
  - ▶ Another way to express perfect matching is an  $N \times N$  (permutation) matrix  $X$ , where  $x_{ij} = 1$  for the connected pair of  $(s_i, b_j)$  and  $x_{ij} = 0$  for all other  $(i, j)$  (call this **assignment**).
- We say that a bipartite graph has a perfect matching if its subgraph is a perfect matching.

## Matching Theorem

There is some nice simple condition that characterizes the existence of a perfect matching for bipartite graph with equal numbers of nodes on the left and right.

Fix bipartite graph  $\langle S \cup B, E \rangle$ . For any set of nodes on one side, say  $B' \subset B$ , let  $N(B') \subset S$  be the set of neighbors of  $B'$  (the set of nodes on the other side that are connected to some node in  $B'$ ). We say that  $B'$  is a **constricted set** if  $N(B')$  contains less nodes than  $B'$ .

### Example 4:



$(b_1, b_2)$  is a constricted set.  $(s_2, s_3)$  is another constricted set.

Note that, for any bipartite graph with the same number of nodes on the left and right, when there is a constricted set on one side, there must be a constricted set on the other side.

## Matching Theorem

It is clear that, if a bipartite graph has a perfect matching, then it cannot have a constricted set.

It turns out that nonexistence of a constricted set is necessary and sufficient for a bipartite graph to admit a perfect matching.

### Matching Theorem (Hall)

A bipartite graph with equal number of nodes on the left and right has a perfect matching if and only if it does not have any constricted set.

The proof is beyond this course.

## Market Clearing Price

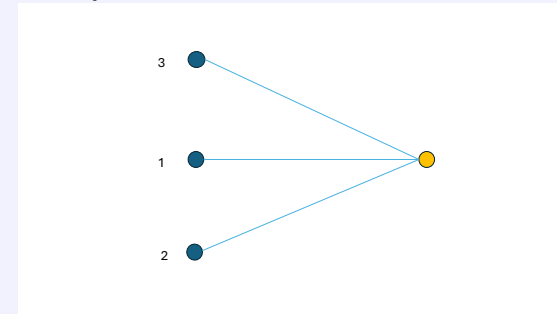
- How are goods traded through a network and what price? And how does the network structure affect them?
- **Market-clearing prices** are prices  $p_i \geq 0, i = 1, \dots, N$  such that there is a balance in the demand and supply of each good. In the current setting, this means that each good is demanded by exactly by one buyer.

Let's define such prices more formally.

- Given price vector  $\mathbf{p} = (p_1, \dots, p_N) \in \mathbb{R}_+^N$ , let seller  $s_i$  be buyer  $b_j$ 's **preferred seller** if  $b_j$ 's **payoff** is maximized (and nonnegative) when she buys from  $s_i$ , i.e.  $v_{ij} - p_i \geq v_{i'j} - p_{i'}$  for any  $i' \neq i$  and  $v_{ij} - p_i \geq 0$ .
- Given  $\mathbf{p}$ , we can draw a **preferred-seller graph** by connecting every buyer to her all preferred sellers.
- $\mathbf{p}$  is a **market-clearing price** if the preferred-seller graph induced by  $\mathbf{p}$  has a perfect matching.
- Let  $X_{\mathbf{p}}$  be the assignment of goods through the perfect matching that is generated by market-clearing price  $\mathbf{p}$ . We call  $(\mathbf{p}, X_{\mathbf{p}})$  a **market equilibrium**.

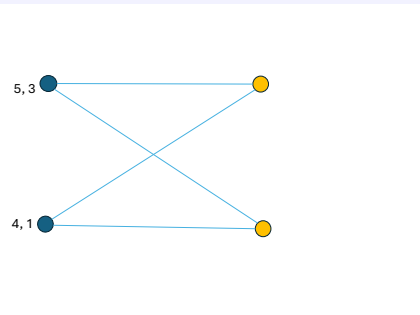
Let's revisit the simple examples to find some examples of market-clearing prices

### Example 1:



The market clearing price must be at least 2 and at most 3. In fact, any price between 2 and 3 is a market clearing price (with 0 price for dummy goods).  $b_1$  is the buyer who wins the good.

### Example 2:



In market equilibrium, buyer  $b_1$  buys good 2 from seller  $s_2$  and buyer  $b_2$  buys good 1 from seller  $s_1$  (we show why this must be true).

Then any  $(p_1^*, p_2^*) \geq 0$  such that  $2 \leq p_1^* - p_2^* \leq 3$  and  $p_1^* \leq 4$  would be market-clearing prices.

## Optimality of Market-Clearing Prices

- Some perfect matchings are more efficient than others.
- Call a perfect matching an **optimal matching** when it maximizes the total sum of values of assigned goods among all possible perfect matchings.
- It can be shown that, if  $\mathbf{p} = (p_1, \dots, p_N)$  is a market clearing price, then the resulting perfect matching must be an optimal matching.

## Optimality of Market-Clearing Prices

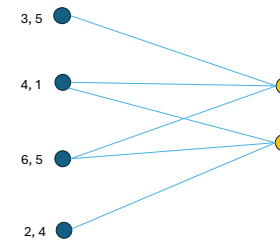
- Take any price  $\mathbf{p} = (p_1, \dots, p_N)$  and any assignment  $X$ . Let  $BS_{X,\mathbf{p}}$  be the total sum of buyers' payoffs given  $\mathbf{p}$  and  $X$  (i.e. buyers' surplus). Let  $V_X$  be the total value (= social surplus) given  $X$ . Then by definition we have

$$BS_X = V_X - \sum_{i=1}^N p_i$$

- Note that assignment  $X$  maximizes the buyers' surplus if and only if it maximizes the total value.
- In the market equilibrium, each buyer maximizes her payoff given price  $\mathbf{p}$ , hence the resulting perfect matching and assignment  $X_{\mathbf{p}}$  must maximize the sum of buyers' payoffs (otherwise...). Hence it must maximize the total value.

Let's find a market equilibrium price and matching for another example.

### Example 3:



In market equilibrium,  $s_1$  must sell good 1 to  $b_3$  and  $s_2$  must sell good 2 to  $b_4$ .

The market-clearing price can be any  $(p_1^*, p_2^*)$  such that  $p_1^* \geq 4$ ,  $p_2^* \leq 4$ , and  $p_1^* - p_2^* \leq 1$ .

## Existence of Market-Clearing Price

- Does a market-clearing price always exist?
- there is an algorithm that always generates a market clearing price (hence it exists).
- We assume that every  $v_{ij}$  take integer values for the algorithm to achieve the market equilibrium exactly.

## Algorithm

- 1 Set  $\mathbf{p}(0) = (0, \dots, 0)$ .
- 2 Draw the preferred graph of buyers given  $\mathbf{p}(t)$ ,  $t = 0, 1, \dots$ . If the graph has a perfect matching, then we are done.
- 3 If not, there must be a constricted set of buyers. Take any minimal such set  $B' \subset B$  (a constricted set that does not contain a smaller constricted set).  
Raise the price of every good sold by  $N(B')$  by 1.
- 4 If the lowest price becomes strictly positive, then reduce every price by 1.  
This becomes the updated price  $\mathbf{p}(t + 1)$ .
- 5 Go to Step 2.

When this algorithm stops, we find market-clearing prices. In fact, it always stops within a finite number of steps:

- Given price  $\mathbf{p}$ , let  $\phi_j^B(\mathbf{p})$  be buyer  $b_j$ 's optimal payoff, let  $\phi_i^S(\mathbf{p}) = p_i$  be seller  $s_i$ 's revenue. Then define potential function by 
$$\phi(\mathbf{p}) = \sum_{j=1}^N \phi_j^B(\mathbf{p}) + \sum_{i=1}^N \phi_i^S(\mathbf{p}).$$
- $\phi(\mathbf{p})$  is **positive**: Starting from 0 prices, the minimum price is always 0. So  $\phi_j^B(\mathbf{p})$  and  $\phi_i^S(\mathbf{p})$  are always positive, hence  $\phi(\mathbf{p})$  is always positive.
- $\phi(p)$  is **decreasing at each step**: In each step, if the algorithm does not stop (i.e. no perfect matching is found), then there is a minimal constricted set  $B'$  with positive potential payoffs. The potential payoff of each such buyer decreases by 1, whereas the potential revenue (=price) of each seller in  $N(B')$  increases by 1. Since  $\#(B') > \#(N(B'))$ ,  $\phi(\mathbf{p})$  must decrease strictly. Note that the adjustment in step 4 does not affect  $\phi(\mathbf{p})$ .

## Multiplicity of Market-Clearing Prices

- The perfect matching that arise in market equilibrium is optimal, hence usually unique. But there are usually multiple market-clearing prices.
- You may guess that the above algorithm is searching for the smallest market-clearing prices. Indeed, it finds the **buyer-optimal** one. If it finds  $\mathbf{p}$ , then for any other market-clearing price  $\mathbf{q}$  we have  $p_i \leq q_i, i = 1, \dots, N$ .
- Of course, then there is **seller-optimal market-clearing price**, which can be found by raising all the prices as high as possible starting from the buyer-optimal one, without disturbing every buyer's incentive to consume the same good (because the assignment must be optimal).

## Example: Online Advertising

- There are two advertising slots. The first one generates 100 clicks per day and the second one generates 50 clicks a day.
- Three buyers competing for these two slots. Each buyer's value per click is  $\theta_1 = 5$ ,  $\theta_2 = 3$ , and  $\theta_3 = 1$ . So, for example,  $v_{21} = 5 \times 50 = 250$ .

What are the market clearing prices and the resulting matching?

- In any market equilibrium, buyer  $i$  gets the  $i$ th slot for  $i = 1, 2$  (why?).
- The minimum possible market-clearing price for the second slot is 50 (otherwise the buyer 3 would demand the second slot). Given this, the minimum possible market clearing-price for the first slot is 200 (otherwise buyer 2 would prefer the first slot to the second slot).
- The algorithm would find these prices (but it is more complicated).
- As for the seller-optimal ones, the maximum possible market-clearing price for the second slot is 150. Given this, the maximum possible market-clearing price for the first slot is 400 (why?).

- More generally, suppose that  $x_1 > x_2 > \dots > x_M > 0$  are on sale and there are  $N$  buyers with valuations per unit  $\theta_1 > \theta_2 > \dots > \theta_N > 0$ . Assume  $N \geq M$ .
- We can show that the  $j$  th buyer must buy  $x_j$  for  $j = 1, \dots, M$  in equilibrium.
- The minimum possible market-clearing price for  $x_M$  is  $p_M = \theta_{M+1}x_M$  for (this price is 0 if  $M = N$ ).
- The minimum possible market-clearing price for  $x_{M-1}$  is  $p_{M-1} = \theta_M(x_{M-1} - x_M) + p_M$ , which follows from the  $M$ th buyer's indifference condition:  $\theta_M x_M - p_M = \theta_M x_{M-1} - p_{M-1}$ .
- ... The prices obtained in this way is indeed the minimum market-clearing price.
- Try to derive the formula for the maximum possible market-clearing prices.

### Example with Non-complete Network:

- Suppose that  $x_1 > x_2 > x_3$  are on sale and there are four buyers with per unit value  $\theta_1 > \theta_2 > \theta_3 > \theta_4$ .
- Assume that  $x_3$  is reserved for buyer 3 and 4. Buyer 1 and 2 are not allowed to buy  $x_3$ . How would this restriction affect the buyer-optimal market-clearing prices and the seller-optimal market-clearing prices?

## Summary

- A simple market model with unit-sellers and unit-buyers with heterogeneous valuations.
- There is always a market-clearing price.
- Any market-clearing price generates the optimal (perfect) matching.
- There are many market-clearing prices. The smallest one is buyer-optimal and the largest one is seller-optimal.
- We find the market-clearing prices for some special markets (online advertising) in a systematic way.

Suggested reading: EK Ch.10.1-10.5, EK Ch.15.1-15.2